

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourgish Professional Newsroom

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form aligns well with deployment needs (LOWER priority dimension). All tasks produce discrete label outputs evaluated by macro-F1 [Q93, Q96], directly compatible with the deployment's per-task significance-score-and-threshold pipeline [A4]. Encoder results are averaged over three runs with reported standard deviations [Q89, Q96], enabling stability assessment. The label-output (not MCQA) paradigm [Q84] mirrors the deployment's classification framing. Class-imbalanced tasks use class-balanced loss [Q92]. Caveats: LLM outputs failing label format are discarded [Q90], inflating reported scores; LLMs are evaluated only once with macro-F1 only [Q96] — meaning LLM performance is less stable, and the benchmark itself flags this as 'indicative rather than directly comparable' [Q86]. The deployment's threshold-flagging logic [A4] requires confidence scores per task, which encoder fine-tuning supports natively but prompted LLM evaluation may not.

ID	Evidence
Q84	"We did not use a Multiple Choice Question Answering (MCQA)-setup, but provided the labels that should be used as output." (p.6)
Q86	"However, prompt-based evaluation introduces additional sources of variability, including prompt sensitivity and differences in instruction-following behaviour across models. As a result, performance should be interpreted as indicative rather than directly comparable to supervised results." (p.6)
Q89	"For encoder-based models, results are reported as averages over multiple runs (see Appendix 7.2 for more details)." (p.7)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q92	"For the supervised models, since the linguistic acceptability and sentiment analysis datasets are highly imbalanced, when fine-tuning on these tasks we use class-balanced loss based on effective size (Cui et al., 2019) with a beta of 0.99." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)

Strengths

- Discrete label outputs evaluated by macro-F1 [Q93, Q96] directly support the deployment's per-task threshold flagging [A4]
- Encoder results averaged over three runs with standard deviations [Q89, Q96] permit stability assessment for production deployment
- Class-balanced loss applied to imbalanced tasks (LA, SA) [Q92] addresses real distributional skew
- Label-output paradigm rather than MCQA [Q84] matches the deployment's classification framing

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Discrete label outputs [Q93, Q148, Q149, Q150, Q151, Q152, Q153] match the deployment's per-task scoring needs [A4]; the deployment's threshold flagging is a downstream wrapper compatible with macro-F1 evaluation.

OF-2: Not applicable — deployment is text-only [deployment_modality]; speech output is not required.

OF-3: Target users are professional journalists/editors with high literacy and trilingual proficiency [WEB-14, WEB-15]; no accessibility constraints introduced by the output form.

OF-4: Minor mismatches: (a) prompted LLM scores are single-run macro-F1 [Q96] less directly comparable to encoder runs [Q86]; (b) discarded malformed LLM outputs [Q90] inflate apparent performance — relevant if the deployment uses prompted LLMs rather than fine-tuned encoders.

ID	Evidence
Q86	"However, prompt-based evaluation introduces additional sources of variability, including prompt sensitivity and differences in instruction-following behaviour across models. As a result, performance should be interpreted as indicative rather than directly comparable to supervised results." (p.6)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ItzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)
Q148	"headline_classification: Decide if the given title/headline fits the text. Output True or False." (p.15)
Q149	"sentiment_analysis: Classify sentiment of the text. Allowed labels: positive, neutral, negative." (p.15)
Q150	"linguistic_acceptability_binary: Decide whether the sentence is linguistically acceptable in Luxembourgish. Output: 0 or 1." (p.15)
Q151	"linguistic_acceptability_multilabel: Detect if the sentence is correct or if some element is wrong. If the sentence is correct, Output: correct. If it is not, Output the label referencing the wrong element: syntax, verb, ortho or adj." (p.15)
Q152	"ner: Perform Named Entity Recognition on the given sequence of sentence tokens. Output tags as lists of ner_tags. Allowed Tags: O, B-LOC, I-LOC, B-PER, I-PER, B-DATE, I-DATE, B-ORG, I-ORG, B-MISC, I-MISC." (p.15)
Q153	"topic_classification: Classify topic of the document by title and text. Allowed category_names: sports, animals, business, culture, technology." (p.15)
WEB-14	Luxembourg trilingual education system produces newsroom professionals with high literacy in Luxembourgish/French/German plus English (https://men.public.lu/en/systeme-educatif/langues-ecole-luxembourgaise.html)
WEB-15	EU Education and Training Monitor confirms strong language education context for target users (https://op.europa.eu/webpub/eac/education-and-training-monitor/en/country-reports/luxembourg.html)

Information Gaps

- Whether confidence scores from fine-tuned encoders are well-calibrated for the deployment's threshold-based flagging logic

Requires Expert Verification

- Calibration of model confidence outputs against editorial flagging thresholds in production